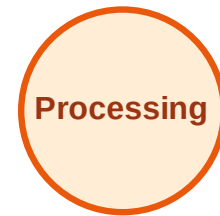


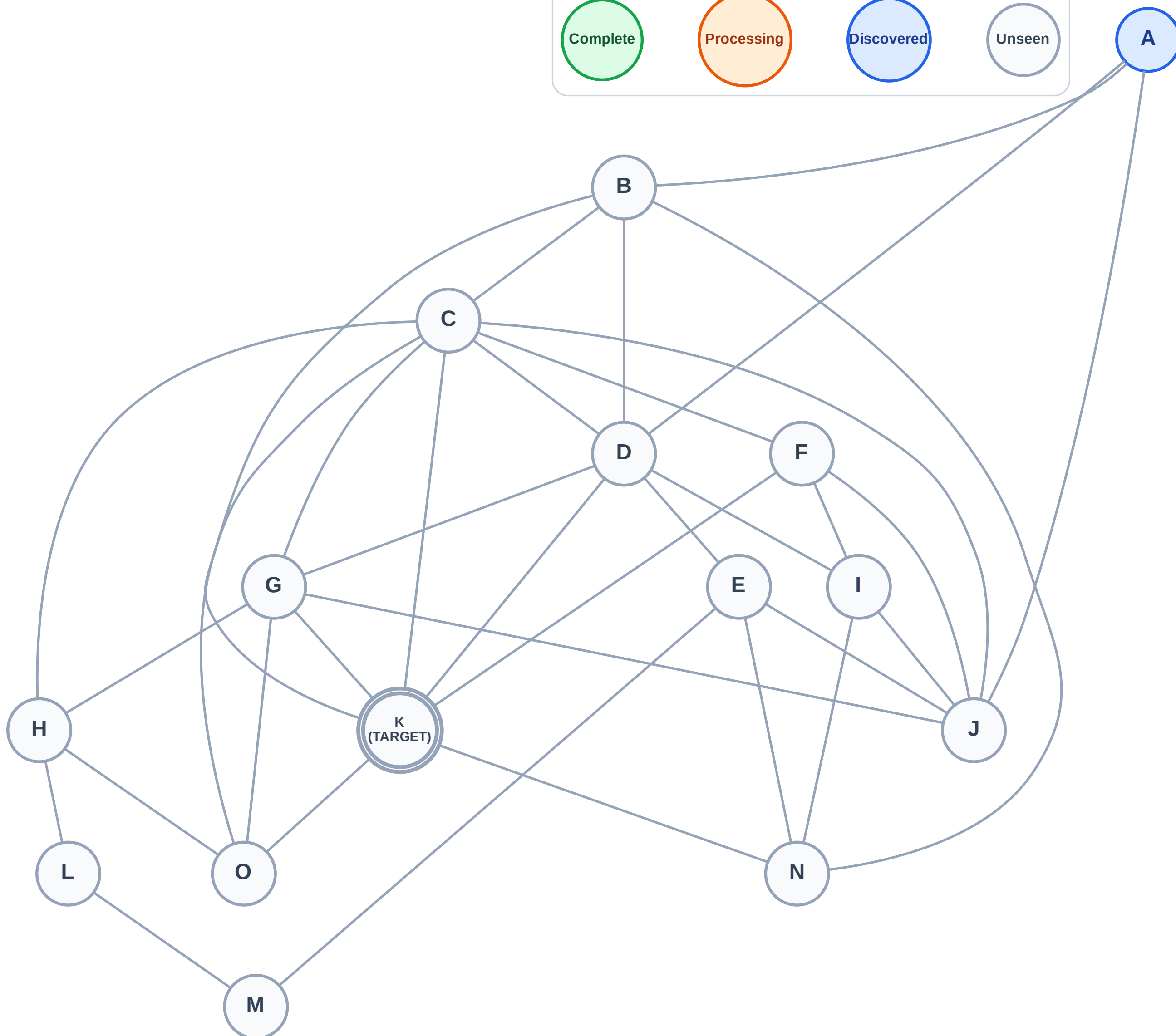
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A
Visited: (none)

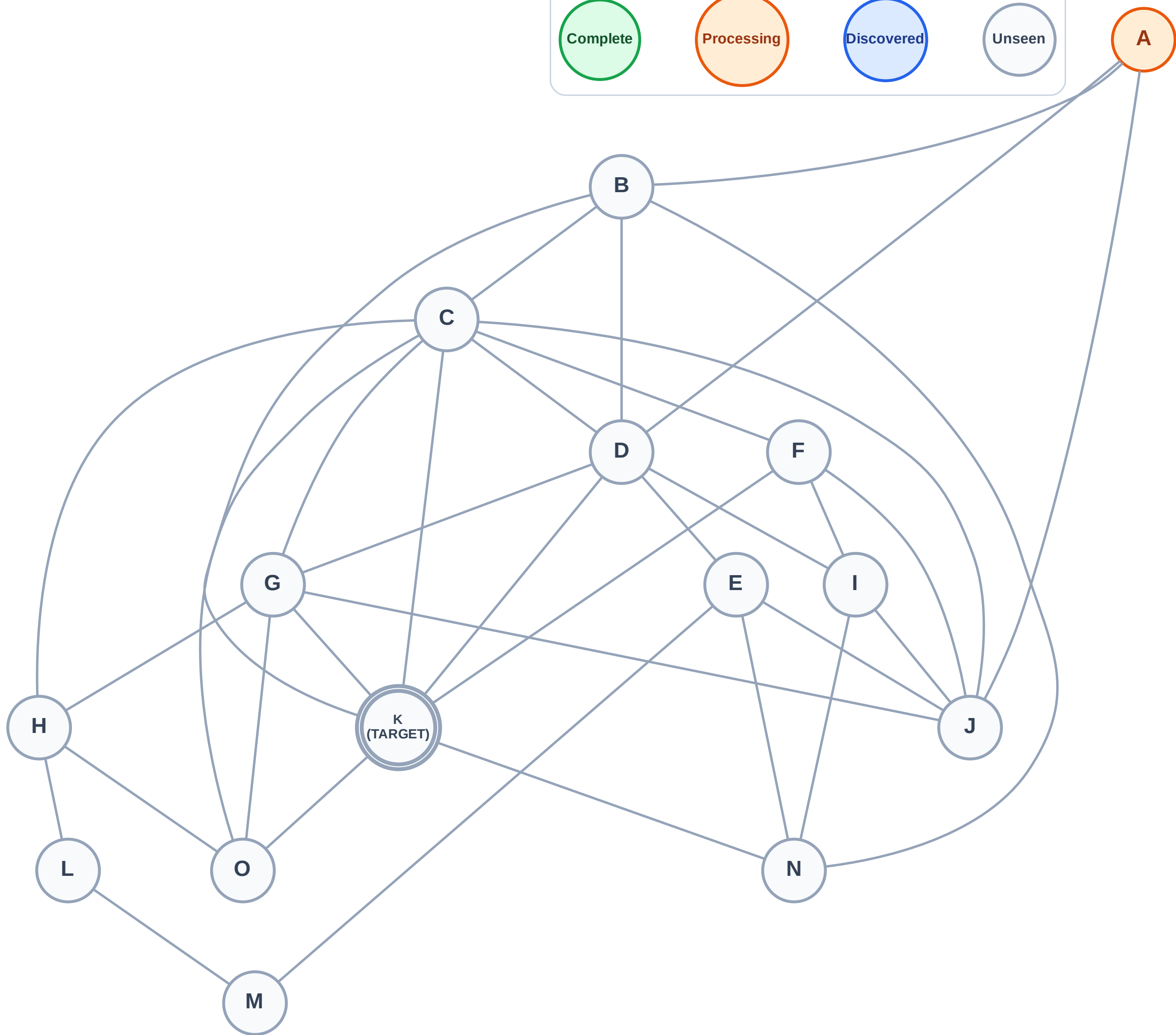


Step 2: Process node A

Legend

Unseen

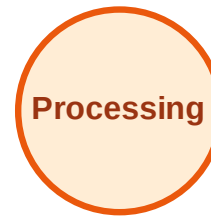
Visited: (none)



BFS Traversal

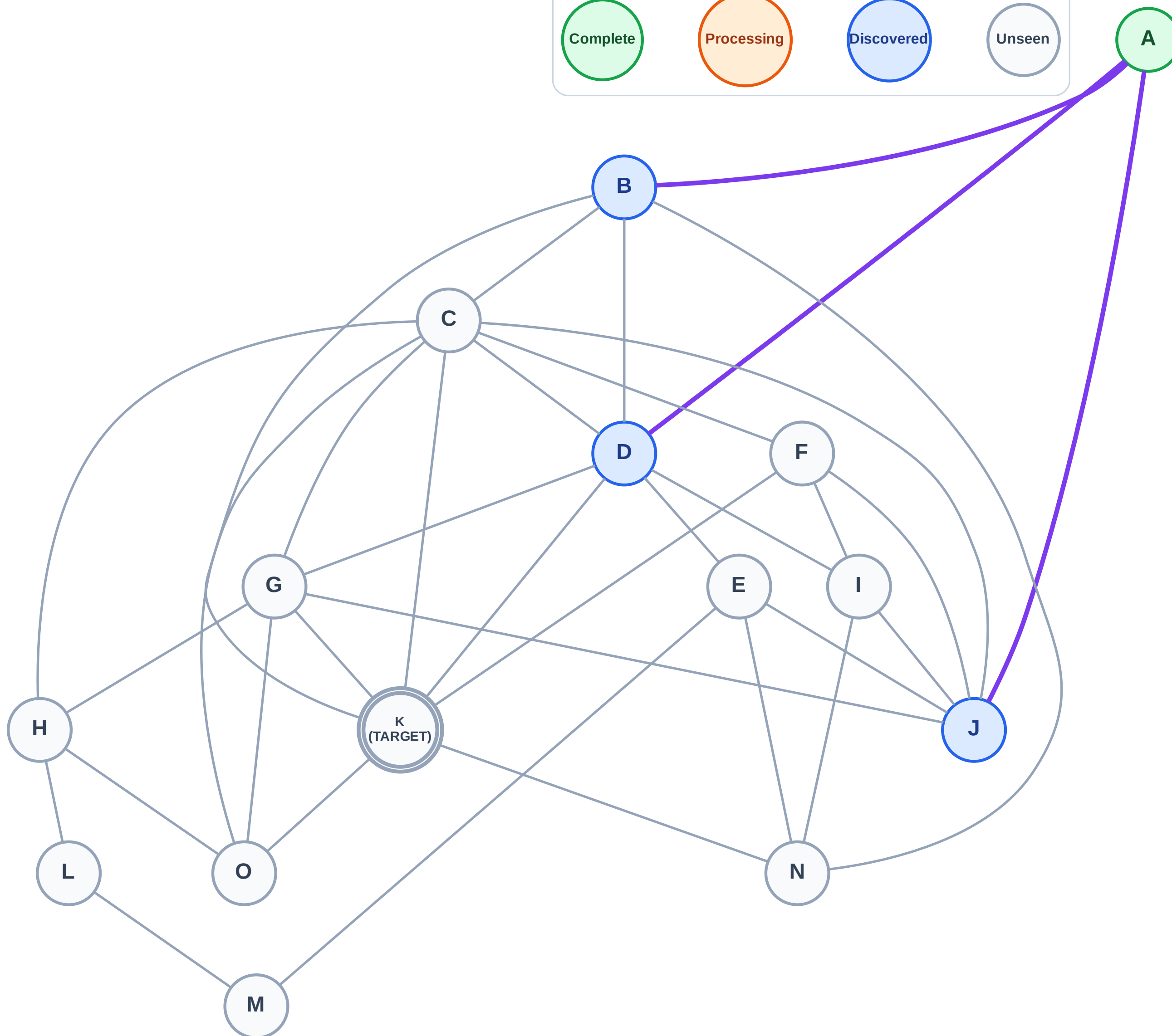
Step 3: Discover B, D, J from A

Legend



Queue: B → D → J

Visited: A



BFS Traversal

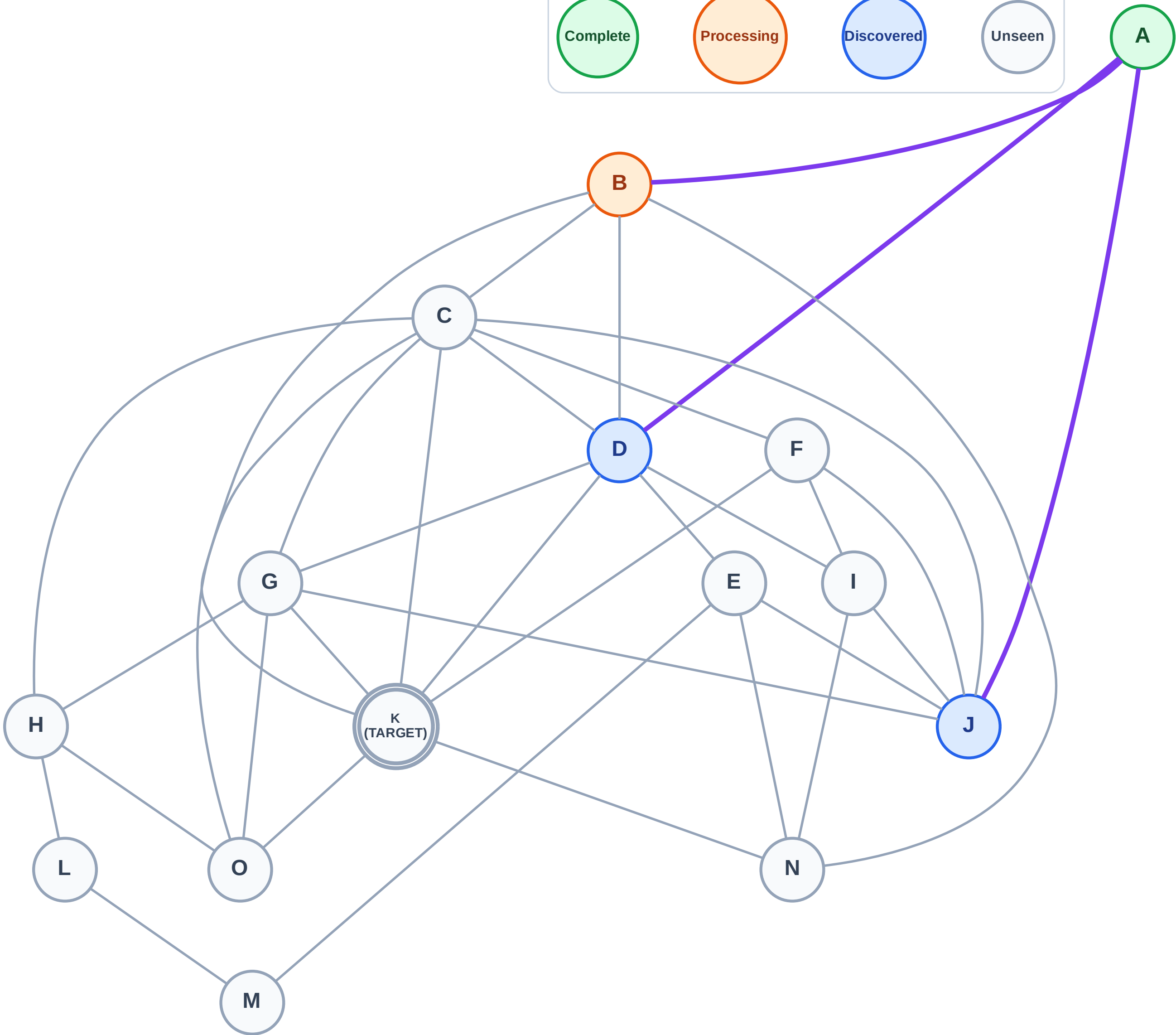
Step 4: Process node B

Legend



Queue: D → J

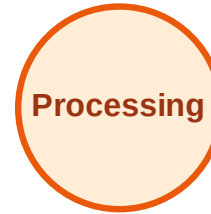
Visited: A



BFS Traversal

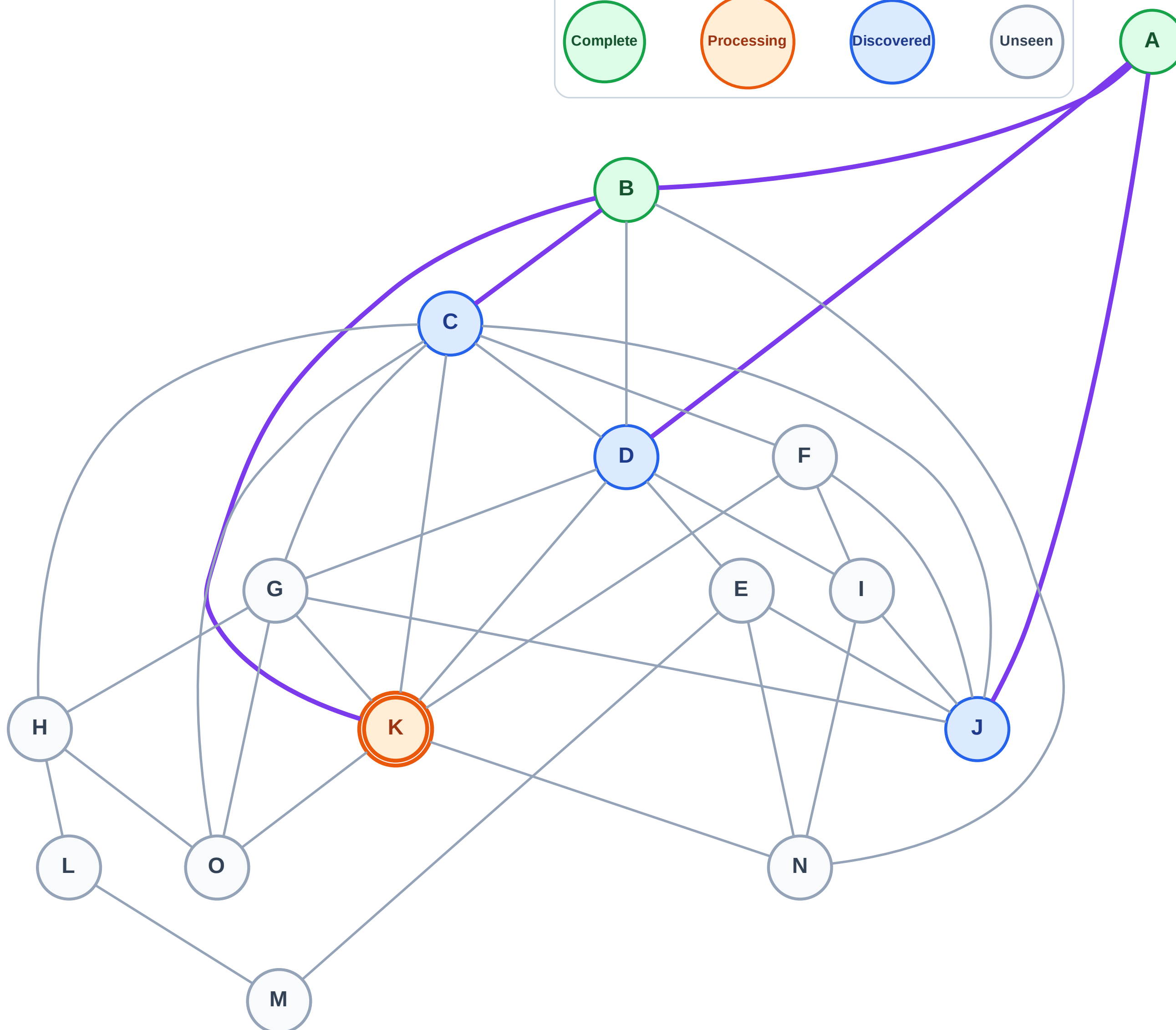
Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Queue: D → J → C

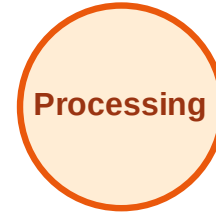
Visited: A, B, K



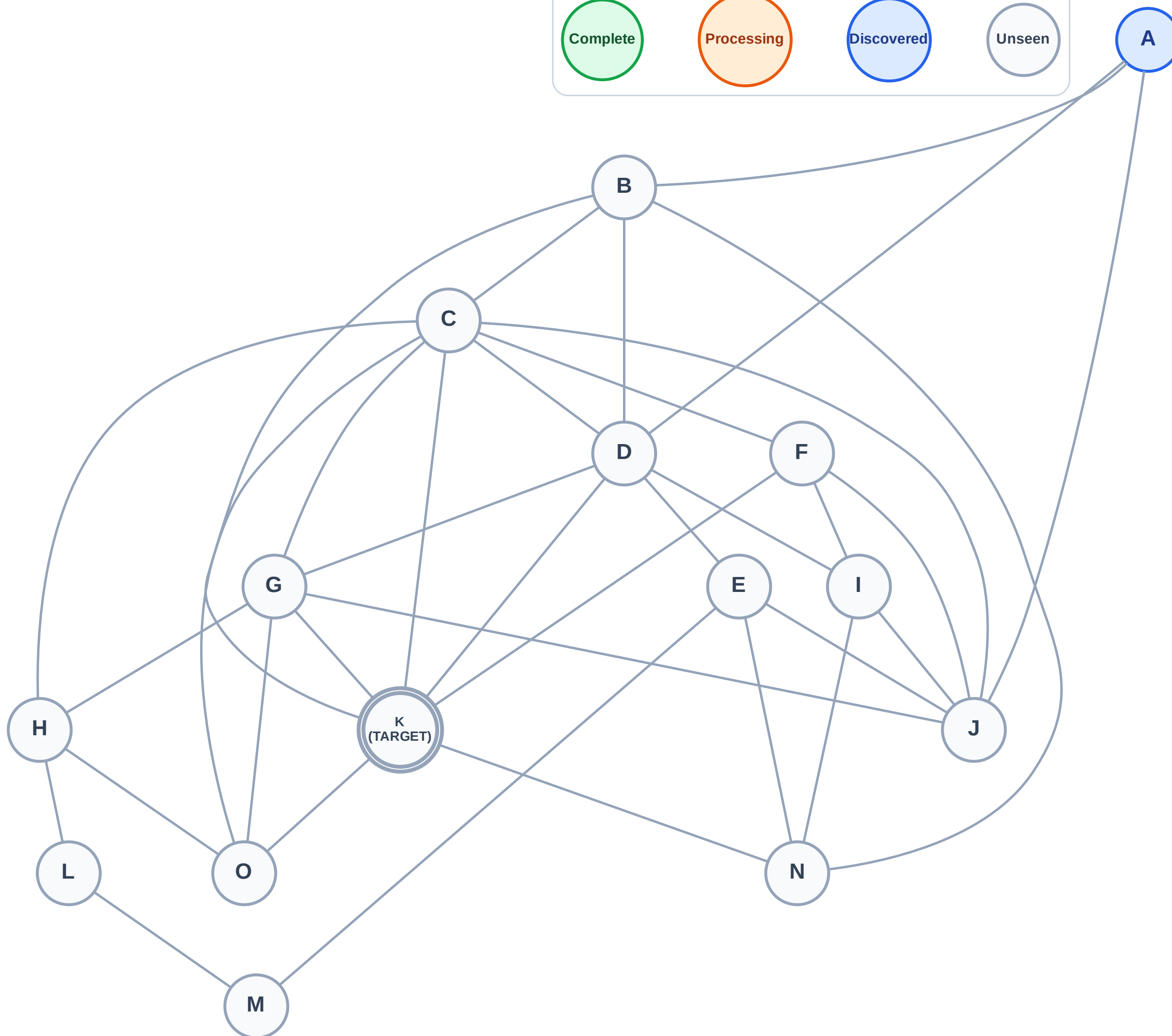
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

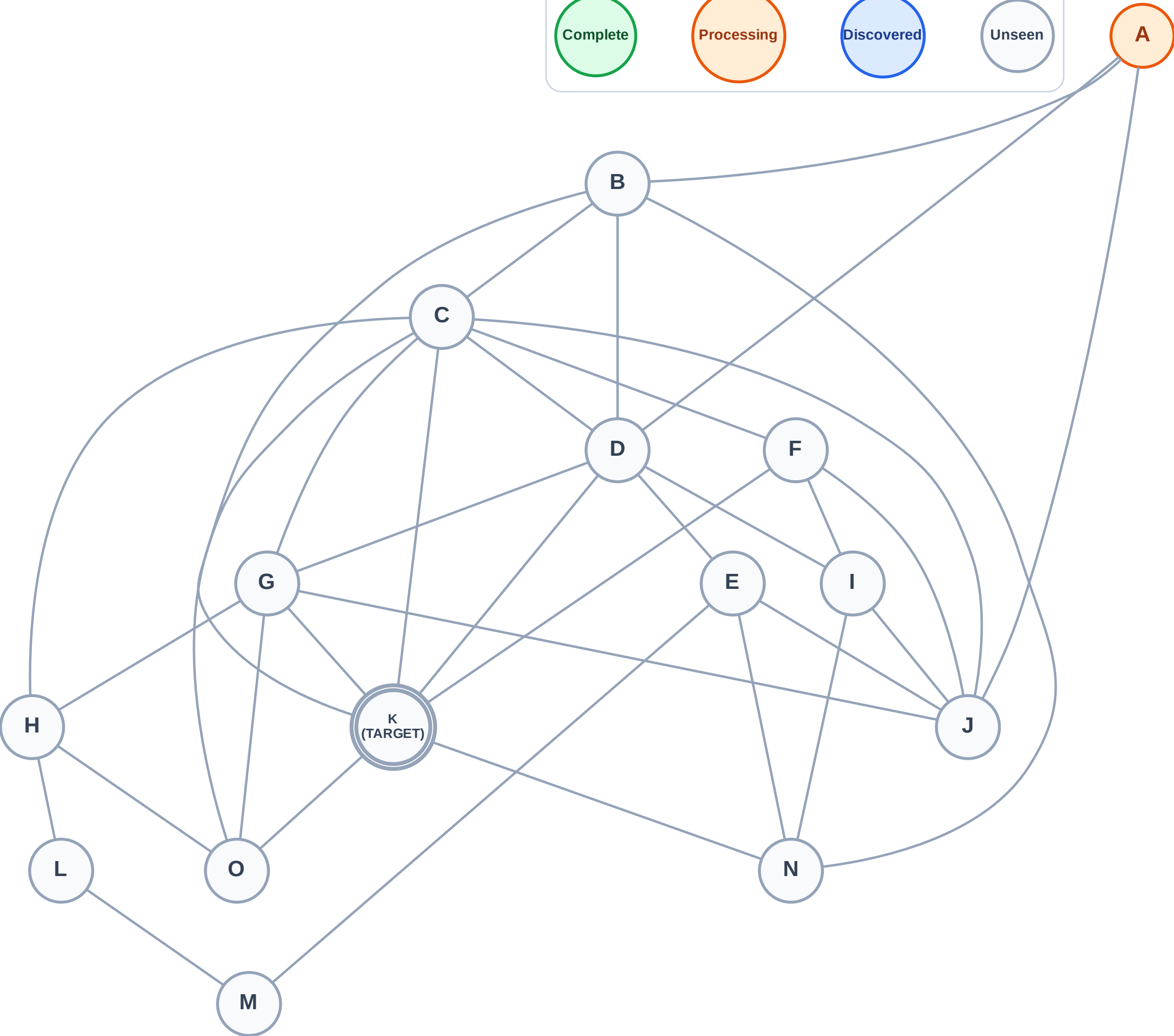
Complete

Processing

Discovered

Unseen

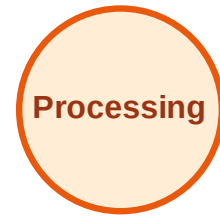
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

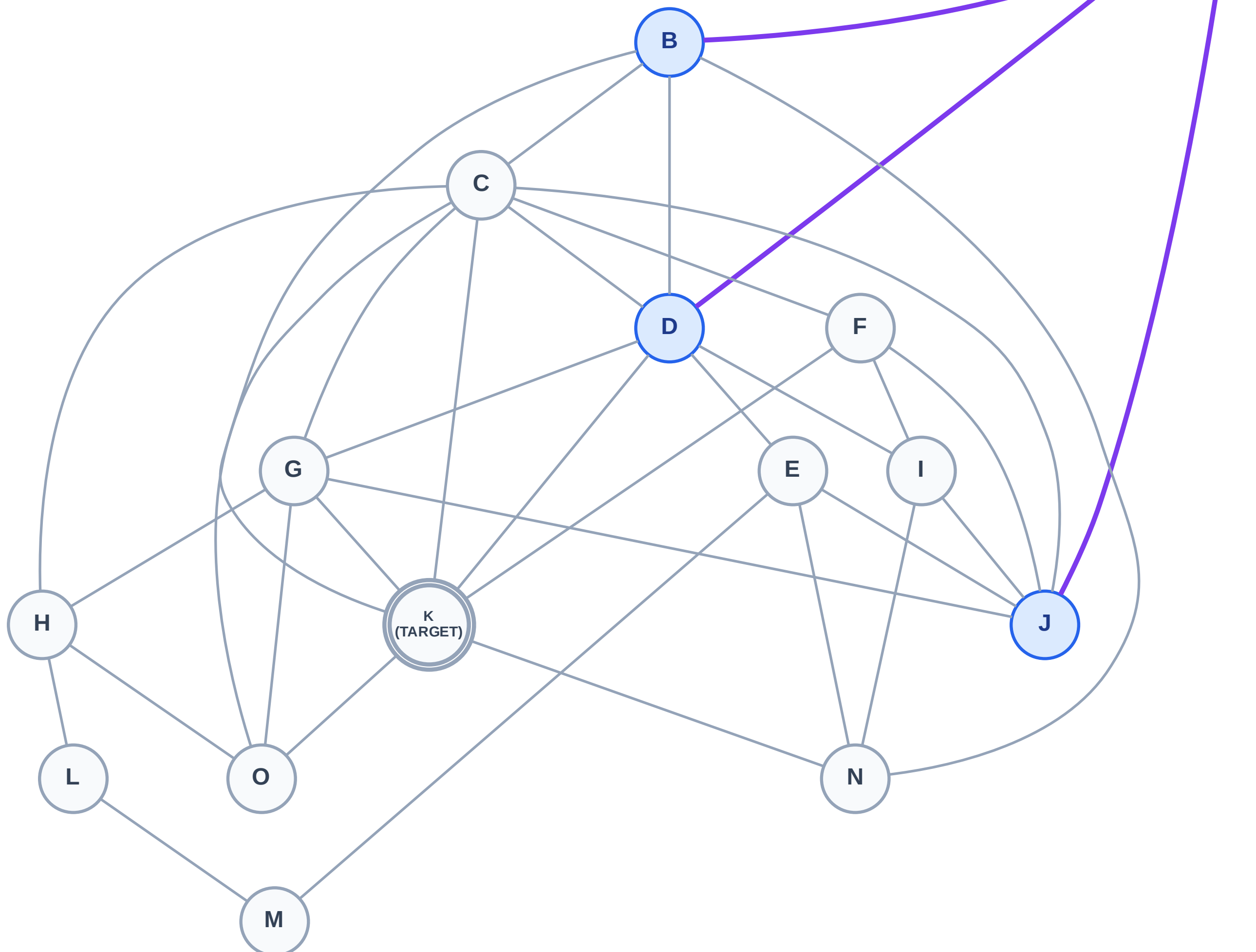
Step 3: Discover J, D, B from A

Legend



Stack (top at right): J | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

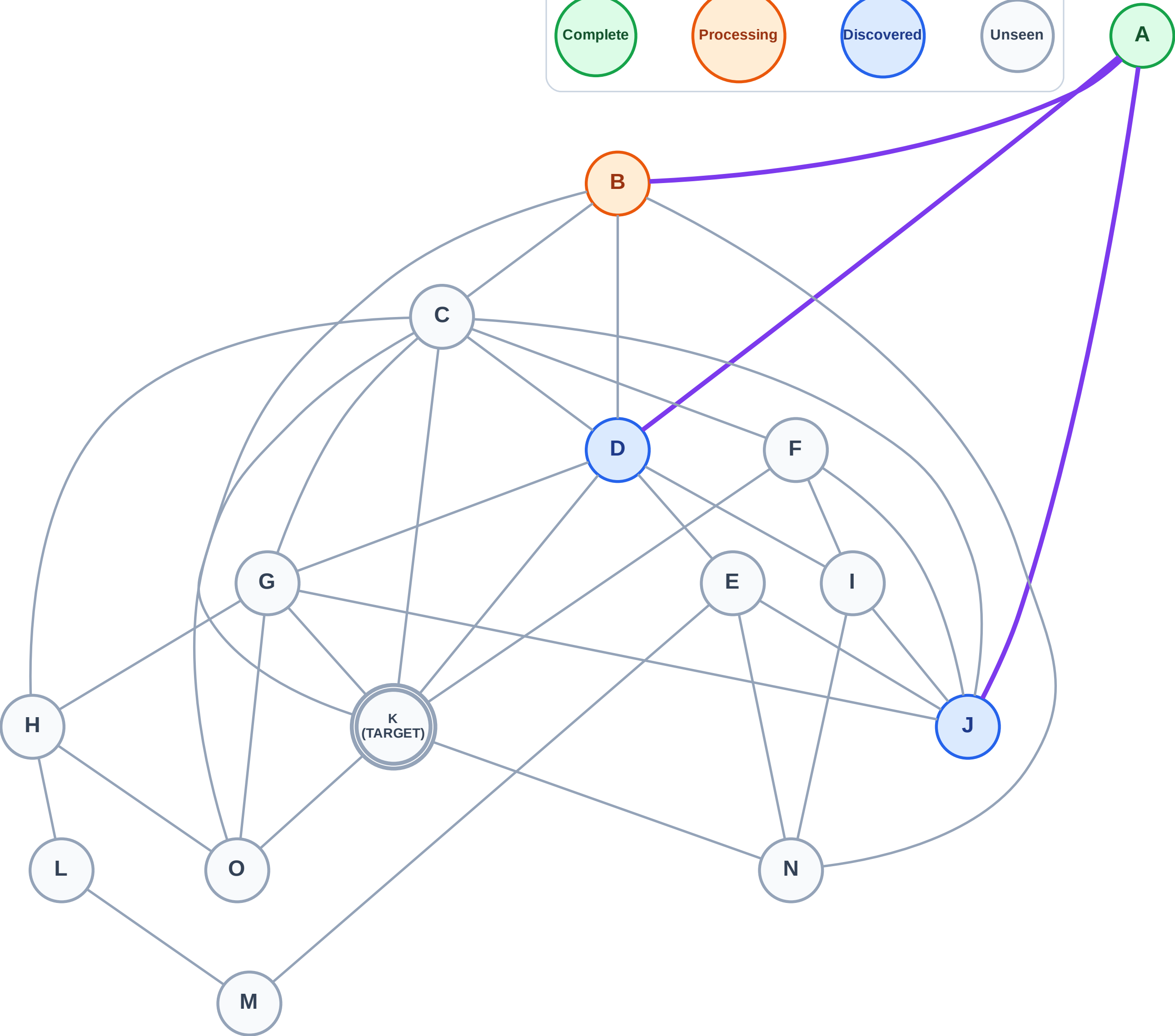
Processing

Discovered

Unseen

Stack (top at right): J | D

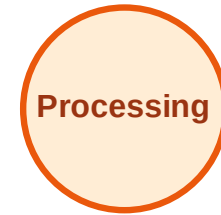
Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Stack (top at right): J | D | N

Visited: A, B, K

